Exp:12 END TO END COMMUNICATION AT TRANSPORT LAYER
29/9/25

(a) Implement ECHO client server using TCP sockets.

AIM:

To implement an ECHO SERVER application using TCP sockets.

Code:

ECHO SERVER

```
import socket
S = socket.socket()
S.bind(("localhost", 5000))
S.listen(1)
C, addr = S.accept()
msg = C.recv(1024).decode()
print("client:", msg)
C.send(msg.encode())
C.close()
S.close()
```

ECHO CLIENT

```
import socket
S = socket.socket()
S.connect(("localhost", 5000))
msg = input("Enter message:")
```

```
s.send(msg.encode())
print("Echo:", s.recv(1024).decode())
s.close()
```

Input:

Enter message: Hello server

Output:

```
Server
Client connected
Received: Hello server
client
Echo from server: Hello server
```

Result:

The Echo client server using TCP sockets was studied and executed successfully.

(b) Implement a chat client server using TCP / IP sockets

Aim:

To implement a chat client server application using TCP sockets.

CODE:

SERVER

```
import socket
s = socket.socket()
s.bind(("localhost", 5000))
s.listen(1)
c, a = s.accept()
while True:
    msg = c.recv(1024).decode()
    print("client:", msg)
    if msg == "bye":
        break
    reply = input("server:")
    c.send(reply.encode())
c.close()
```

CLIENT

```
import socket
s = socket.socket()
s.connect(("localhost", 5000))
while True:
```

```
msg = input("client :")
s.send(msg.encode())
if msg == "bye":
    break
print("server:", s.recv(1024).
                    decode())
s.close().
```

INPUT

client : Hello
server : Hi
client : Bye

OUTPUT

client : Hello
server : Hi

Result:

The chat client server applications, were successfully studied and implemented using TCP sockets.